

Revised Generation Interconnection System Impact Study Report

For Queue Project Q22-2034

South Colton 138 kV
March 01, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2034. The Interconnection Customer proposes a 76.5 MW battery energy storage facility interconnecting at the South Colton 138 kV substation in the Virginia area. The request was received on March 16, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 709 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 21 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,222,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Monel / La Joie 138.0 kV	New Lodon / Franklin-Elk Grove 138.0 kV	153.6	212	8.9
Clover / Leesville 345.0 kV	State School / Clover Substation 345.0 kV	124.7	1,643	17.8
Intel / S Crk Substation 138.0 kV	S Crk / Aibonito Pds 138.0 kV line	123.8	404	9.2
Lyndell Substation / Orlando Cogen Lp 138.0 kV	S Crk / Wythe 138.0 kV Line	109.1	299	5.4
Riv Substation / Layfir 138.0 kV	Wythe / South Creek 138.0 kV Line	108.9	274	5.4
Riv Substation / Perth Substation 138.0 kV	Wythe / South Creek 138.0 kV line	108.8	281	5.4
Orlando Cogen Lp / Perth 138.0 kV	South Crk / Wythe 138.0 kV Line	108.6	287	5.4
Nacoochee / S Texa 138.0 kV	Flowood / S Colton 138.0 kV	106.9	659	5.5
Desert Outpost / S Texa 138.0 kV	S Texa / Franklin-Elk Grove 138.0 kV Line	106.0	1,002	8.2
Franklin-Elk Grove / S Texa Substation 138.0 kV L	Desert Outpost Substation / S Texa Substation 138.0 kV	105.7	774	8.4
Wythe / South Creek 138.0 kV	New London / Franklin-Elk Grove 138.0 kV	105.4	629	13.4
Walworth / Shenandoah 138.0 kV L	Bear / Clover 345.0 kV line	105.2	520	5.5
Red Hill / Grassmere 138.0 kV	Bear / Fluvanna P.S Substation 345.0 kV	105.1	472	5.6
Shenandoah / Aibonito Pds 138.0 kV line	Bear / Clover Substation 345.0 kV	105.0	544	5.5
Franklin-Elk Grove / Banning 138.0 kV Line	Franklin-Elk Grove / South Texa 138.0 kV	104.9	509	10.9

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Yaphank / S Colton 138.0 kV	Desert outpost / S Texa 138.0 kV L	104.8	517	17.6
Desert Outpost / Banning 138.0 kV	Franklin-Elk Grove / South Texa 138.0 kV	104.5	523	10.9
Person / State School 345.0 kV	Person / Mecklenburg Power Station Substation 345.0 kV	102.3	2,134	15.3
Wythe / Unity 138.0 kV	Parkin / Opossum Crk 138.0 kV	102.1	368	13.5
Person / Riverview 345.0 kV Line	Leesville / Clover 345.0 kV	102.0	1,969	21.8
South Creek / Aibonito Pds 138.0 kV	Bear / Clover 345.0 kV line	100.2	799	8.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Monel / La Joie 138.0 kV	Rebuild line with higher-capacity conductor	\$20,413,000	3% (\$699,000)
Leesville / Clover 345.0 kV	Rebuild line with higher-capacity conductor	\$128,610,000	1% (\$1,365,000)
Intel / S Crk 138.0 kV line	Rebuild line with higher-capacity conductor	\$9,468,000	3% (\$243,000)
Lyondell / Orlando Cogen Lp Substation 138.0 kV	Reconductor line	\$6,497,000	6% (\$362,000)
Riv Substation / Layfir 138.0 kV	Reconductor line	\$1,146,000	6% (\$64,000)
Riv Substation / Perth Substation 138.0 kV	Reconductor line	\$2,010,000	6% (\$112,000)
Orlando Cogen Lp / Perth 138.0 kV	Reconductor line	\$4,169,000	6% (\$232,000)
S Texa / Nacoochee 138.0 kV	Reconductor line	\$1,191,000	1% (\$11,000)
South Texa Substation / Desert Outpost 138.0 kV Line	Reconductor line	\$790,000	1% (\$8,000)
Franklin-Elk Grove / S Texa 138.0 kV	Reconductor line	\$4,169,000	1% (\$53,000)
South Crk / Wythe 138.0 kV	Reconductor line	\$4,268,000	2% (\$99,000)
Shenandoah / Walworth 138.0 kV line	Reconductor line	\$8,394,000	4% (\$370,000)
Grassmere / Red Hill 138.0 kV	Reconductor line	\$998,000	3% (\$32,000)
Shenandoah / Aibonito Pds 138.0 kV line	Reconductor line	\$10,861,000	4% (\$479,000)
Banning / Franklin-Elk Grove 138.0 kV	Reconductor line	\$520,000	2% (\$10,000)
Yaphank / S Colton 138.0 kV	Reconductor line	\$1,907,000	6% (\$111,000)
Banning / Desert Outpost 138.0 kV	Reconductor line	\$5,108,000	2% (\$103,000)
Person / State School 345.0 kV	Reconductor line	\$18,318,000	1% (\$185,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Wythe / Unity 138.0 kV	Reconductor line	\$4,174,000	10% (\$406,000)
Person / Riverview 345.0 kV Line	Reconductor line	\$28,594,000	1% (\$237,000)
South Creek / Aibonito Pds 138.0 kV	Reconductor line	\$2,214,000	2% (\$41,000)
	Total allocated to project		\$5,222,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.